



Fox School of Business

TEMPLE UNIVERSITY®

Department of Risk, Actuarial, Healthcare Management, Legal Studies

AS 3504/5108

Actuarial Analytics

1. General Course Information

<u>Course Name:</u>	AS 3504/5108; Actuarial Analytics
<u>Meeting Time and Place:</u>	MW 9:30-10:45am in Alter 607
<u>Homework Access:</u>	Coaching Actuaries Adapt (All Links on Canvas)
<u>Required Materials:</u>	Frees, E. W. (2009). <i>Regression modeling with actuarial and financial applications</i> . Cambridge University Press. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). <i>An introduction to statistical learning: with applications in R</i> . New York: Springer.
<u>Lecture Notebook:</u>	I have found having a notebook (or digital equivalent) each class for taking notes, doing problems, quizzes/other assignments helps students better absorb the material.
<u>Calculator:</u>	SOA approved calculator (TI-30XS Multiview is best)
<u>Credit Hours:</u>	3 Credits
<u>Course Description:</u>	This course introduces students to statistical learning and linear models, with a focus on applying these tools to actuarial business decisions in an insurance or consulting environment. Topics to be covered include: types of modeling problems, methods of assessing model accuracy, exploratory data analysis, exponential family of distributions, parameter estimation, diagnostic tests of model fit and assumptions, model selection and interpretation, calculation of predicted values and confidence/prediction intervals in the context of ordinary least squares regression, k-nearest neighbors, and generalized linear models. The course aims to prepare students for the data analytics responsibilities of actuaries through its discussion of statistical techniques and actuarial applications of predictive analytics, and development of programming skills.
<u>Course Prerequisite:</u>	AS 2505 Actuarial Statistics or STAT 2512 with a minimum grade of C
<u>UEC:</u>	Completion of both this course and AS4504 Advanced Actuarial Analytics above the 85% threshold determined by the Society of Actuaries will grant University Earned Credit for SOA Exam SRM.



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2. Instructor Information

Instructor / Professor Name: **Jeffrey Zheng (he/him)**

E - Mail Address: jzheng@temple.edu (Best)

I value getting back to students within 24 hours during the weekdays. If you haven't heard from me, I'd appreciate a follow-up email as I would not want to miss your message.

Canvas:

I post all the course content on Canvas. Please take the time to read through the announcements and module resources to best support your learning.

Office Hours:

Mon, Wed 2-3pm

Tue 12:30-1:30pm

Thu 10-10:50am

Fri (Zoom 1on1 Meetings upon request)

Please contact me if you can't make any of the regularly scheduled, as I care about being accessible to all my students.

Office Room:

Alter Hall A613

Zoom ID: 7555971915

About Me:

I have always loved working with numbers and interacting with people and found a way to merge my passions through teaching actuarial science at Temple. I worked as a property & casualty actuary at CNA insurance for 7 years prior to academia. I care deeply about student learning and career success, so I am always eager to interact with you and get to know you as unique individuals. I am a proud Temple Owl and completed my EdD dissertation on determinants of college student graduation.

3. Course Learning Objectives

Students will:

- A. Explain the key concepts of statistical learning and differentiate types of modeling problems and methods.
- B. Describe and contrast the components of linear models and generalized linear models.
- C. Estimate parameters using least squares and maximum likelihood and calculate predictions and intervals.
- D. Interpret diagnostic tests of model fits and assumptions.



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- E. Justify the selection of an appropriate model based on statistical tools and application context and interpret model results.

4. Grading

Grading Scale:

A [93%-100%], **A-** [90%-92.9%],
B+ [87%-89.9%], **B** [83%-86.9%], **B-** [80%-82.9%],
C+ [77%-79.9%], **C** [73%-76.9%], **C-** [70%-72.9%],
D [60%-69.9%], **F** [0%-59.9%]

Table 1 Grade Determination:

Component	Temple Grade Weight	UEC Grade Weight
Attendance, Participation	10%	0%
Weekly Homework	25%	15%
Semester Project	10%	5%
Exam #1	15%	15%
Exam #2	15%	15%
Final Exam (Cumulative)	25%	50%
Gamma Extra Credit	Max +3% on Final	None

UEC Requirements: UEC credit may be granted to those who score at least above 85% overall using the UEC grading scale in BOTH AS3504 & AS4504. All exams are in-person, proctored, closed notes and closed book, with an SOA approved calculator. SOA formula sheets are provided. Exams cannot be dropped or retaken. Make-ups in emergency situations are only provided within one week of the original exam and must include documentation in accordance with SOA and Temple University policy.

You must receive a grade of at least a “C” in this course to take Advanced Actuarial Analytics as your elective.

Late Policy: 1 Late Pass on any homework for any reason, 10% off per day otherwise unless documented issue

Grade Feedback Policy: Any individual grade questions should be asked via personal appointment outside of class within 1 week of receiving the grade.



I am willing to provide feedback throughout the semester so students can see how they are doing and identify places to apply more effort or new strategies. My hope is all of you grow your content mastery and improve throughout the semester through practice. Any student's grades should not be tied to how others perform, so there is no grading on a curve in this course.

5. Attendance & Participation Policy

I believe that attending class provides social benefits and community building beyond academic learning. Attendance will be taken when class begins each period. I understand that life happens and I am always willing to accommodate and excuse absences. Unexcused absences do impact your Attendance/Participation grade. All I ask is that you communicate with me in advance of missing class when one of the following or a similar instance occurs. I will remember to Zoom record the lecture.

- Family caregiving / obligations
- Illness / Death
- Required legal proceedings
- Other school-life conflicts

Participation will be holistically graded on the entirety of the semester. It encompasses asking questions, answering questions, answering on polls, actively working actively during in-class examples, emailing me/our peer tutor with questions, coming to my/our peer tutor's office hours, etc. There are many different ways you can engage with the class, even if you are introverted.

I commit to a safe learning environment where diverse perspectives are recognized and valued. I ask that all of you help me work towards a class culture of open communication, mutual respect, and inclusion. Personal attacks will not be tolerated as we strive to build an open and welcoming classroom together. Please keep your cell phones silent and away during class out of respect for the entire class.

6. Homework Policy

Homework is meant to help you learn. I expect you to do your Coaching sets to mastery (15+ correct).

Grading Rubric (Max 15):

Completion to # Correct: +9



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Accuracy 75% or Above: +6

Accuracy at least 65% but less than 75%: +5

Accuracy at least 55% but less than 65%: +4

Accuracy at least 45% but less than 55%: +3

Accuracy at least 35% but less than 45%: +2

Accuracy at least 25% but less than 35%: +1

Accuracy less than 25%: 0/15

If any homework score is not 100%, I expect to see you at office hours.

As a 3-credit course, you can budget about 6 hours a week on average of work outside of class.

7. Exams Policy

Exam dates are listed in the syllabus and you will be reminded of them in prior classes. The final exam is during finals week; the date is set by the University.

General Format

Types of Questions: Free-response quantitative problems, True/False conceptual questions with explanation, and Multiple Choice

All examinations are closed book and closed notes. You may not bring any other written information with you for use during the exam. Doing so constitutes cheating.

I am interested in your learning and your approach to problem solving. Partial credit will be given on all exams, so please show your work.

Materials Covered in Exams

Lecture notes, class discussions/lecture, homework and R assignments

Calculators

Only [SOA approved calculators](#) are permitted for exams. Use of any other calculators constitutes cheating.

Materials Allowed during Exams

Exam, Paper/Pencil/Pen, Approved Calculator



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- No electronic devices including smart watches, should be on your person, on your clothes, on your desk or in your work area during exams. All should be stored in your book bag, purse, etc. and off.
- The presence or use of any type of communication/electronic device during an examination or quiz constitutes cheating. If this happens, you will be required to hand in your exam at that time and to leave the classroom. At minimum, I reserve the right to give a zero in the exam if this occurs and reserve the right to give you an F in the course.
- If you believe that you deserve points for an answer that was graded as incorrect please see me in office hours within one week after you receive the graded assignment, exam etc.
- I want students to put their best foot forward. If you have 3 or more exams on any of the scheduled exam days, please let me know at least 1 week in advance, and I will work with you to reschedule to another date.
- You are all capable of meeting my expectations. If you are concerned about how you're doing in this course, please come talk with me instead of considering academic misconduct.

UEC Exam Make-Up Guidelines

1. If a student misses an exam, for that exam to count for UEC grading, they must take the approved exam within seven days of the original exam date. Official documentation for missing the exam must be provided.
2. After seven days, students may still take the approved exam for the course grade but only in very rare cases will that exam count for UEC grading. This will be subject to the SOA External Examiner's approval.
3. If the exam is taken substantially later than the original exam date, students could shift the exam weight to the final exam weight. According to UEC rules, students will lose potential UEC credit eligibility.

8. Temple University Guidelines

Academic Code of Honor

"As a member of the Temple community, I will not participate in or tolerate academic dishonesty."

[Temple Student Rights](#)

[Temple Student Responsibilities](#)

A breach of the academic policy of Temple University regarding plagiarism or academic



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cheating will be considered grounds for failing this course. Students are directed to the following link of the Undergraduate Bulletin regarding Plagiarism and Academic Cheating. I strongly encourage you to read [this policy](#).

Temple University is committed to the inclusion of students with disabilities and provides accessible instruction, including accessible technology and instructional materials. The process for request access and accommodations for this course is: (1) Advise me of the need for access or accommodations; (2) Contact Disability Resources and Services to request accommodations; (3) DRS will consult with me as needed about essential components of the program; (4) Present me with a DRS accommodation letter. Please read through these policies on Student Academic Rights and Responsibilities.

Inclement Weather Policy:

Please be advised that while Temple University campuses may close for inclement weather, online courses are not on-campus and therefore are still expected to meet. Your instructor will contact you regarding any adjustments needed in the event of a power outage or severe circumstances. Should you have any questions, please contact the professor.

Academic Honesty:

Learning is both an individual and a cooperative undertaking. Asking for and giving help freely in all appropriate setting helps you to learn. You should represent only your own work as your own. Personal integrity is the basis for intellectual and academic integrity. Academic integrity is the basis for academic freedom and the University's position of influence and trust in our society. University and school rules and standards define and prohibit "academic misconduct" by all members of the academic community including students. You are asked and expected to be familiar with these standards and to abide by them.

[Temple's Policy on Academic Dishonesty](#)

Disability Statement:

Any student who has a need for accommodations based on the impact of a documented disability or medical condition should contact Disability Resources and Services (DRS) located in the Howard Gittis Student Center South, 4th Floor at drs@temple.edu or 215-204-1280 to request accommodations and learn more about the resources available to



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you. If you have a DRS accommodation letter to share with me, or you would like to discuss your accommodations, please contact me as soon as practical. I will work with you and with DRS to coordinate reasonable accommodations for all students with documented disabilities. All discussions related to your accommodations will be confidential.

[Disability Resources](#)

[Land Acknowledgement](#)

[Temple University's Technology Usage Policy](#)

Temple University Resources:

Temple provides extensive supports for students to help all of you achieve your full academic and career goals. Virtually all students access these resources at some point in their Temple journey.

[Cherry Pantry](#)

Laptop – If you do not own a laptop, or if something happens during the semester to your laptop, you can borrow a laptop from Temple's [Laptop Share Program](#) (where you can borrow a laptop for up to four hours at no cost). You can request a free, long-term laptop loan for the current or future semester through the [Student Technology Assistance Application](#). You can also rent a laptop in weekly, monthly and semester-long intervals through [OWLtech](#).

[Center for Student Professional Development](#)

[Student Success Center](#)

[Tuttleman Counseling](#)

[Undergraduate Research Support](#)

[University Libraries](#)



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9. Course Outline:

Table 2 Weekly Agenda:

Week of	Topics
1: 8/24	Class Introductions (0.5 Class) Statistical Learning Concepts (1.5 Classes) Canvas Syllabus Quiz (Due Fri) Recap HW: Questions to me after every complete topic Text: Frees Ch. 1, James Ch. 2
2: 8/31	Statistical Learning Recap (0.5 Class) Simple Linear Regression (1.5 Classes) Text: Frees Ch. 2.1-2.3, James Ch. 3.1
3: 9/9 (1 Class)	Simple Linear Regression (0.5 Class) Simple Linear Inferences (0.5 Class) Text: Frees Ch. 2, James Ch. 3.1
4: 9/14	Simple Linear Regression Recap (0.5 Class) Simple Linear Inferences (1.5 Classes)
5: 9/21	Simple Linear Inferences Recap (0.5 Class) Multiple Linear Regression (1.5 Classes) Text: Frees Ch. 3, James Ch. 3
6: 9/28	Multiple Linear Regression (0.5 Class) Exam #1 Review (0.5 class) Exam #1 (Wed)
7: 10/5	Multiple Linear Regression Recap (0.5 Class) MLR Inferences (1.5 Classes) Text: Frees Ch. 3.4, 6
8: 10/12	MLR Inferences (0.5 Class) Linear Model Assumptions (1.5 Class) Text: Frees Ch. 5, James Ch. 3.3
9: 10/19	MLR Inferences Recap (0.5 Class) Linear Model Assumptions (0.5 Class) Model Selection (1 Class) Text: Frees Ch. 5, James Ch. 5, 6.1, 6.4
10: 10/26	Linear Model Assumptions Recap (0.5 Class) Model Selection (1 Classes) Generalized Linear Models (0.5 Class) Text: Frees Ch. 13
11: 11/2	Model Selection Recap (0.5 Class) Exam #2 Review (0.5 Class)



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	Exam #2 (Wed) Work on your group project!
12: 11/9	Generalized Linear Models (2 Classes) Text: Frees Ch. 13 Work on your group project!
13: 11/16	Generalized Linear Models Recap (0.5 Class) Logistic & Probit Models (1.5 Classes) Text: Frees Ch. 11 Work on your group project!
Break:	Have a happy and safe Thanksgiving!
14: 11/30	Logistic & Probit Models (0.5 Class) Poisson Count Models (1.5 Classes) Text: Frees Ch. 12
15: 12/7 (1 Class)	CA: 9 – Logistic, Probit Models (Due Mon) Logistic & Probit Models Recap (0.5 Class) Poisson Count Models (0.5 Class) I will answer Poisson Model Qs on Final Exam Discussion Semester Project (Due Mon) CA: 10 – Poisson Models (Due Fri)
Final: Wed 12/9 8-10am	Cumulative Exam